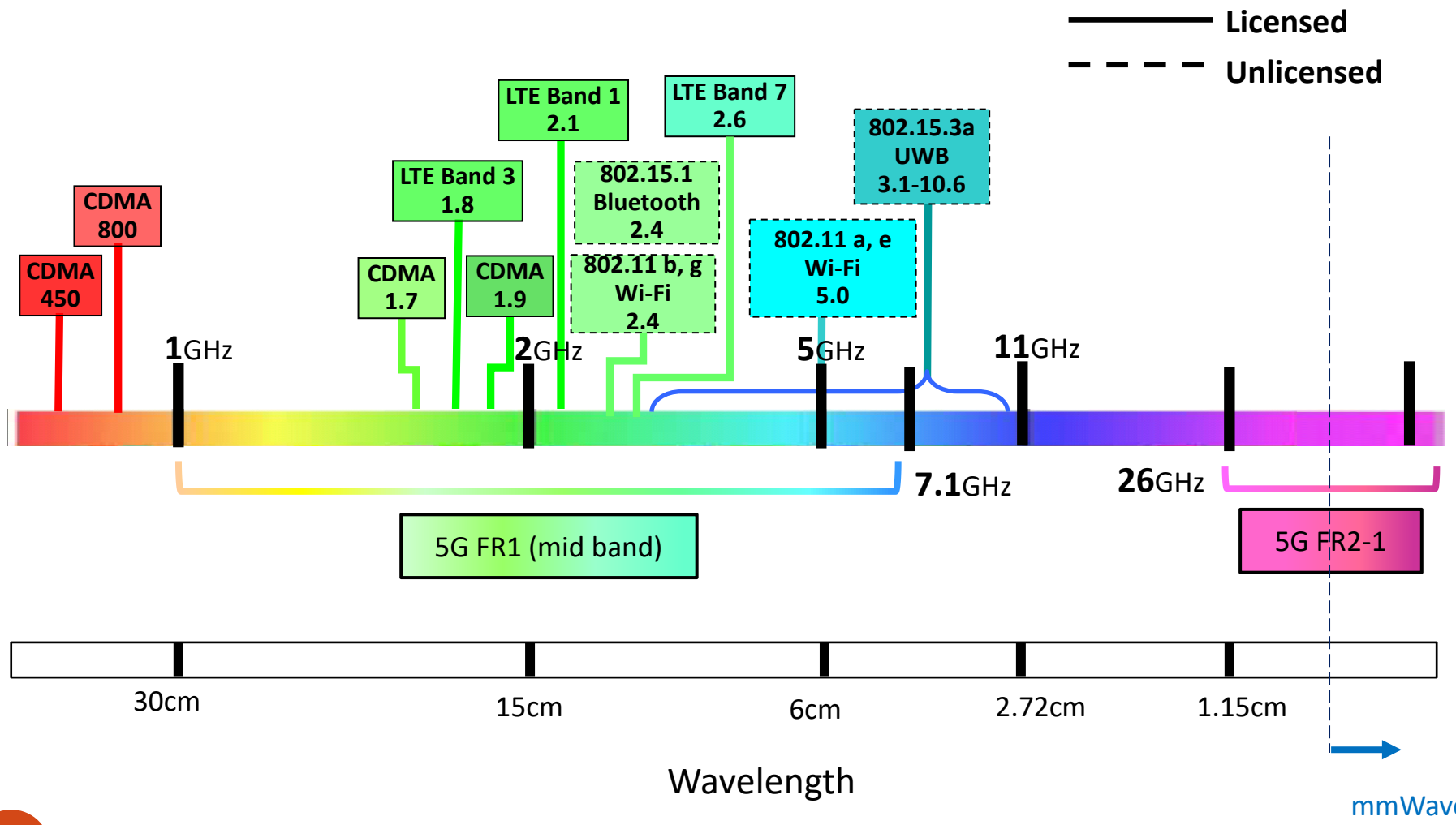
Digital Modulation

Instructor : Pei-Yun Tsai

Digital Communication IC Design

"Baseband Receiver Design for Wireless MIMO-OFDM Communications" 2nd Edition, Wiley 2012.
Chapter 2, Digital Modulation

Spectrum


Licensed
Unlicensed
CDMA 450
CDMA 800
LTE Band 3 1.8
CDMA 1.7
CDMA 1.9
LTE Band 1 2.1
802.15.1 Bluetooth 2.4
802.11 b, g Wi-Fi 2.4
LTE Band 7 2.6
802.11 a, e Wi-Fi 5.0
802.15.3a UWB 3.1-10.6
1GHz
2GHz
5GHz
11GHz
7.1GHz
26GHz
5G FR1 (mid band)
5G FR2-1
30cm
15cm
6cm
2.72cm
1.15cm
Wavelength
mmWave

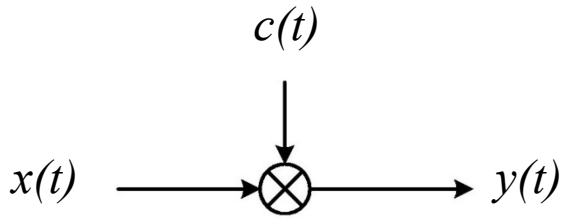
Modulation

- Fourier transform of a time-domain signal *x(t)*

$$X(f) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

- Inverse Fourier transform

$$x(t) = \mathcal{F}^{-1}[X(f)] = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$



$$y(t) = c(t)x(t)$$

$$Y(f) = C(f) \otimes X(f)$$

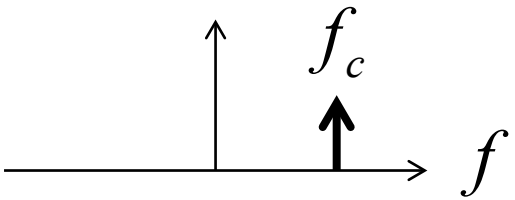
Time Domain v.s. Frequency Domain

- Time domain

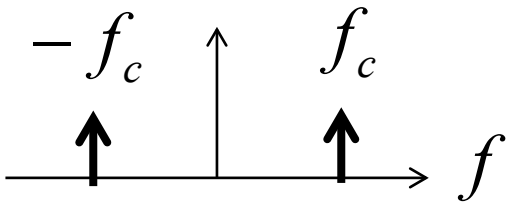
$$c_1(t) = e^{j2\pi f_c t}$$

$$c_2(t) = \cos(2\pi f_c t)$$

- Frequency domain

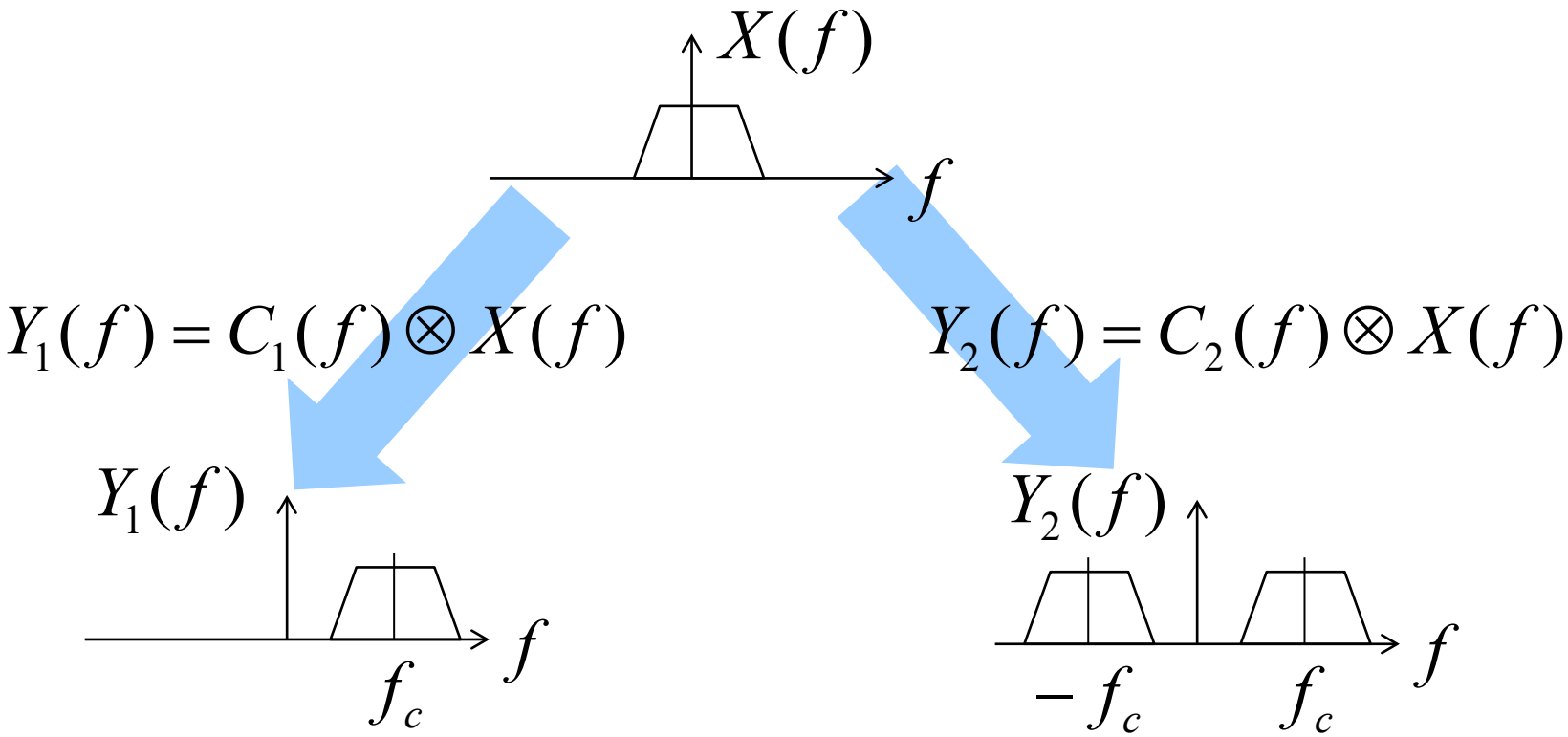


$$C_1(f) = \delta(f - f_c)$$



$$C_2(f) = \frac{1}{2}(\delta(f - f_c) + \delta(f + f_c))$$

Spectrum



Why Modulation?

- Spectrum utilization
- Parameters of the sinusoidal wave to be changed
  - Amplitude
  - Phase
  - Frequency
- Digital modulation
  - One segment of waveforms represents a symbol.
  - A symbol may carry information of several bits.

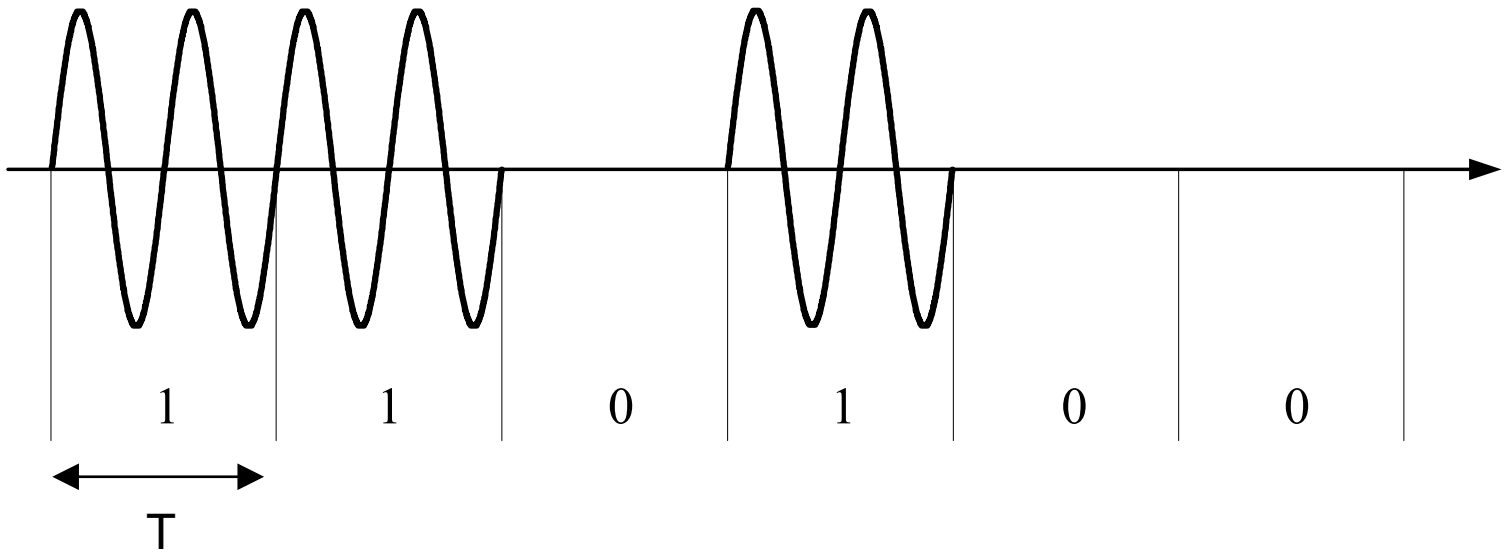
Digital Modulation

- Single-carrier modulation
  - ASK
  - FSK
  - BPSK/QPSK/MPSK
  - DQPSK/OQPSK/(π/4)-DQPSK
  - QAM
- Multi-carrier modulation
  - Orthogonal frequency-division multiplexing (OFDM)
- Two-dimensional modulation
  - Orthogonal time-frequency space modulation (OTFS)
- Spatial modulation

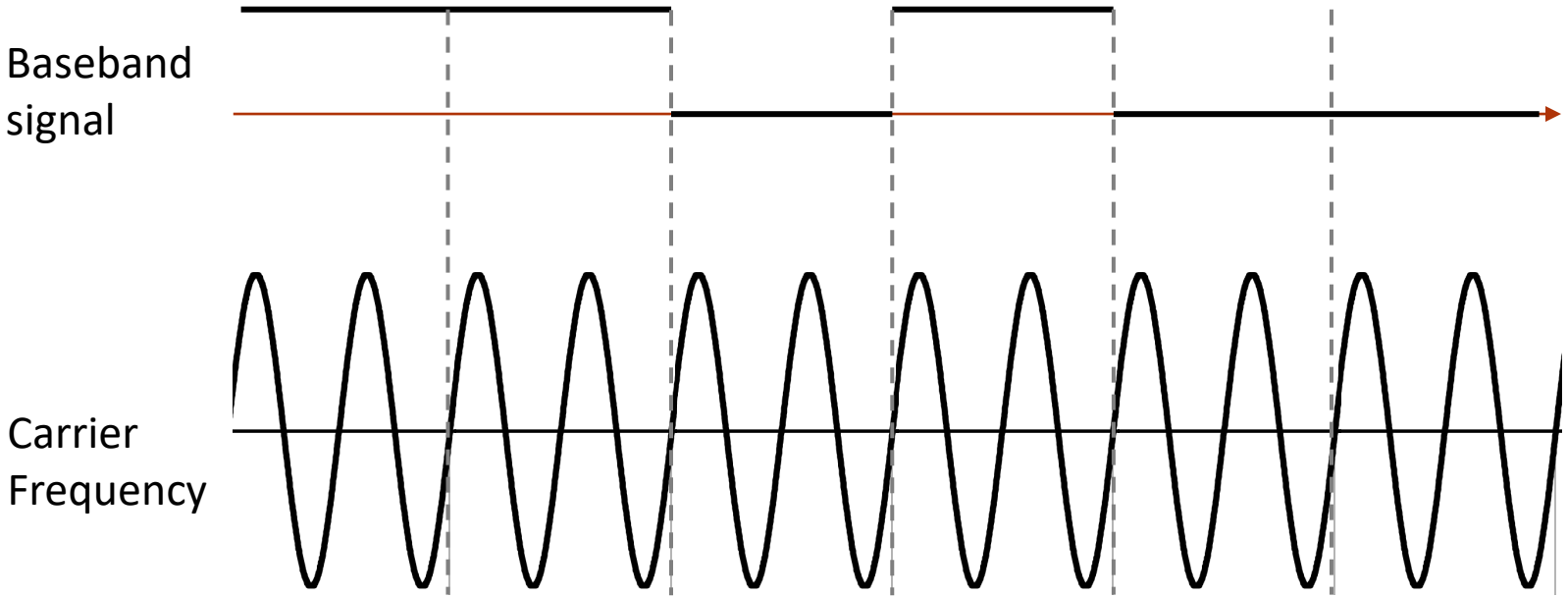
Single-Carrier Modulation

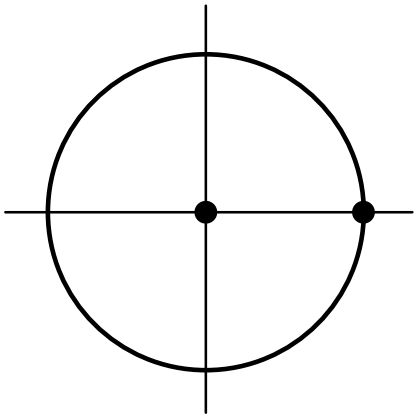
Amplitude Shift Keying (ASK)

- Binary ASK $x(t) = A_k \cdot \cos(2\pi f_c t - \theta)$ $kT \leq t < (k+1)T$



Let's Separate Baseband Signal





$$A_k = \begin{cases} 1 & b_k = 1 \\ 0 & b_k = 0 \end{cases}$$

Baseband Carrier

$$x(t) = A_k \cdot \cos(2\pi f_c t - \theta) = Re\{A_k e^{j(2\pi f_c t - \theta)}\}$$

$$kT \le t < (k+1)T$$

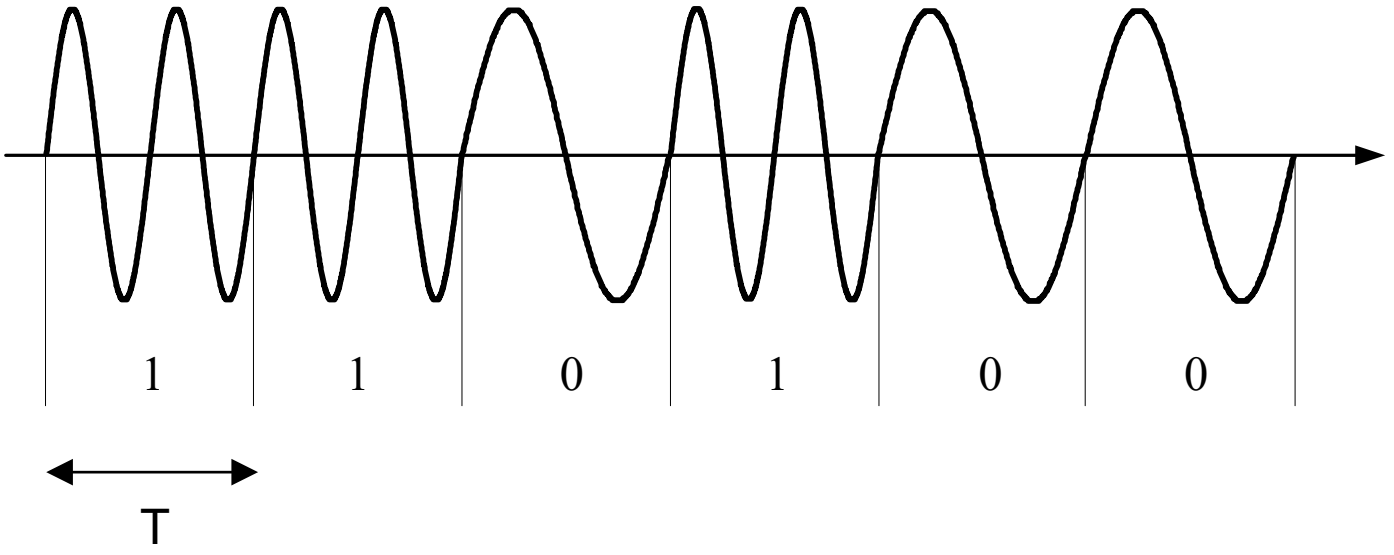
FSK

- Symbols consist of an interval of sinusoidal wave with different frequencies.
- Baseband waveform (data to be transmitted) determine which frequency is used.

Frequency Shift Keying (FSK)

- Two different carrier frequencies to carry one bit.

$$x(t) = \cos(2\pi(f_c \pm \Delta f_k)t - \theta) \qquad kT \le t < (k+1)T$$



If We Separate Baseband Signal

$$x(t) = \cos(2\pi(f_c \pm \Delta f_k)t - \theta) = Re\{e^{j2\pi\Delta f_k t} e^{j(2\pi f_c t - \theta)}\}$$

$$\Delta f_k = \begin{cases} \frac{1}{3} f_c & b_k = 1 \\ -\frac{1}{3} f_c & b_k = 0 \end{cases}$$

BPSK/QPSK/MPSK

- Symbols consist of an interval of sinusoidal waves with different phases but same frequency and amplitude.

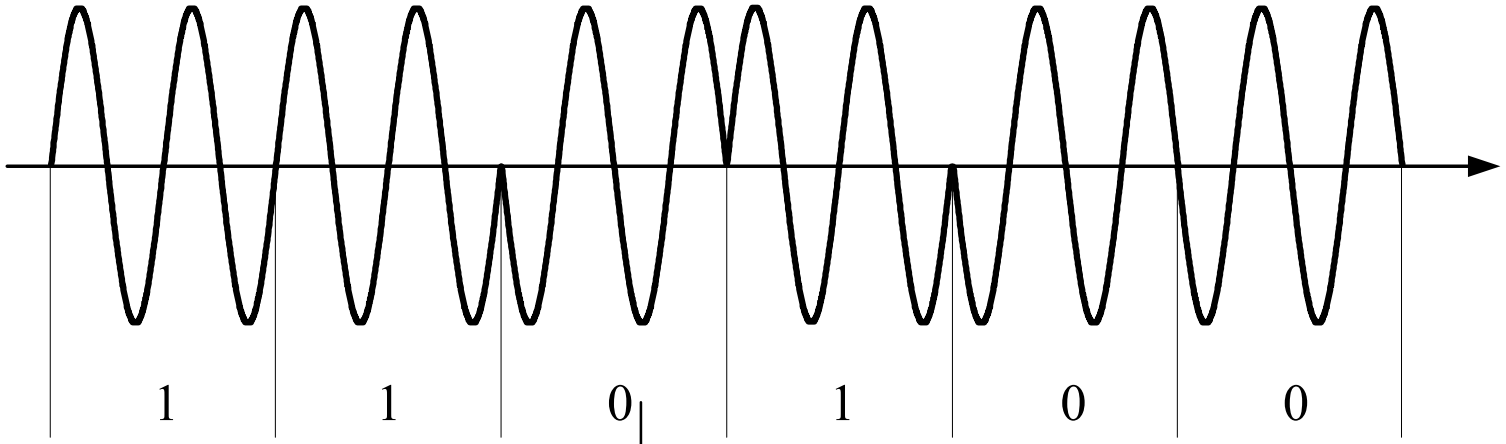
Binary Phase Shift Keying (BPSK)

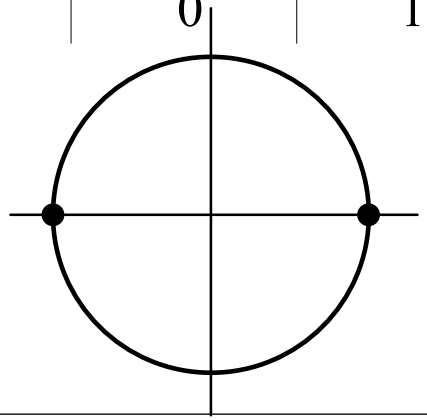
- One single carrier, two different phases.

baseband

passband

$$x(t) = \cos(2\pi f_c t \pm \phi_k - \theta) = Re\{e^{j(2\pi f_c t - \theta)} e^{\pm j\phi_k}\} \qquad kT \le t < (k+1)T$$



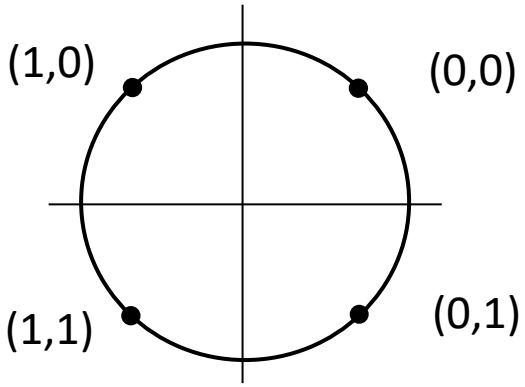


Quaternary Phase Shift Keying (QPSK)

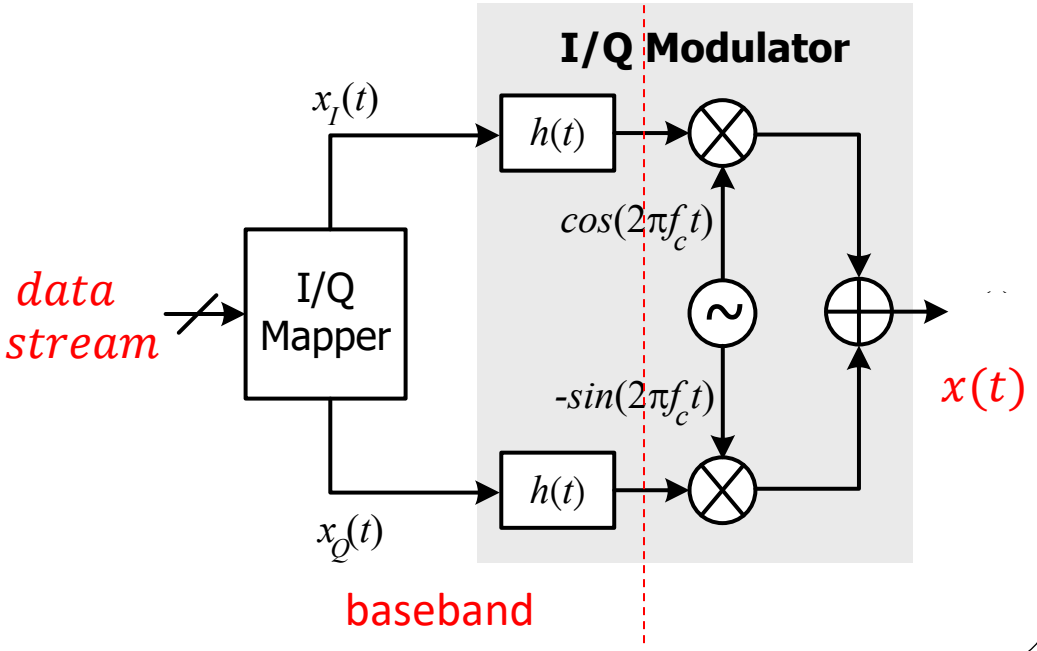
- Four different phases to carry two bits.

$$x(t) = \cos(2\pi f_c t + \phi_k) = Re\{e^{j2\pi f_c t} e^{j\phi_k}\}$$
$$= Re\{\big(\cos(2\pi f_c t) + j sin(2\pi f_c t)\big) x_{BB}(t)\}$$
$$= Re\{\big(\cos(2\pi f_c t) + j sin(2\pi f_c t)\big)(x_{\mathrm{I}}(t) + j x_Q(t))\}$$

$$kT \le t < (k+1)T$$
$$x_{BB}(t) = \sum_{k=0}^{\infty} e^{j\phi_k} h(t - KT)$$



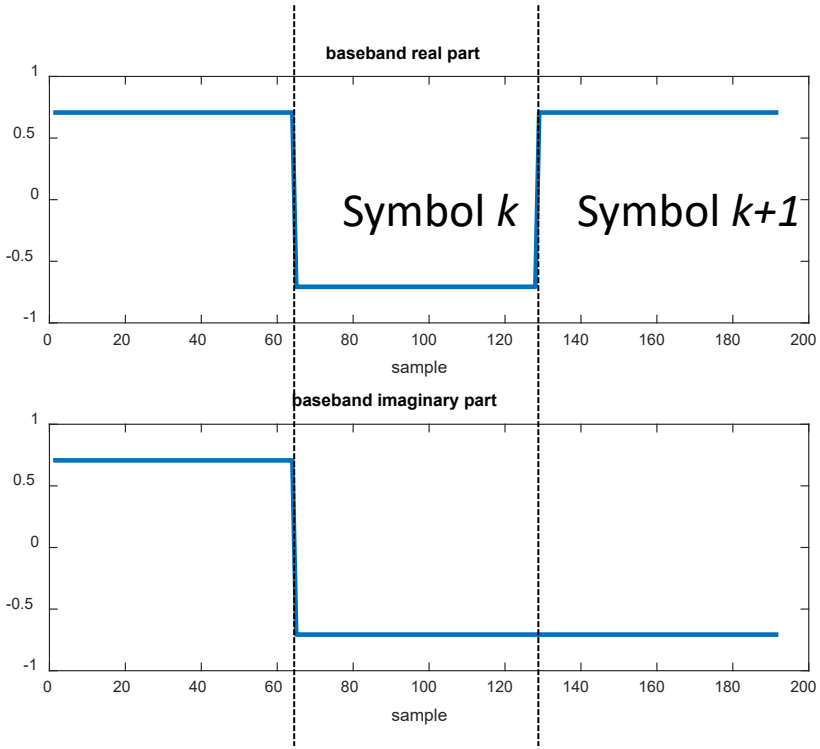
baseband

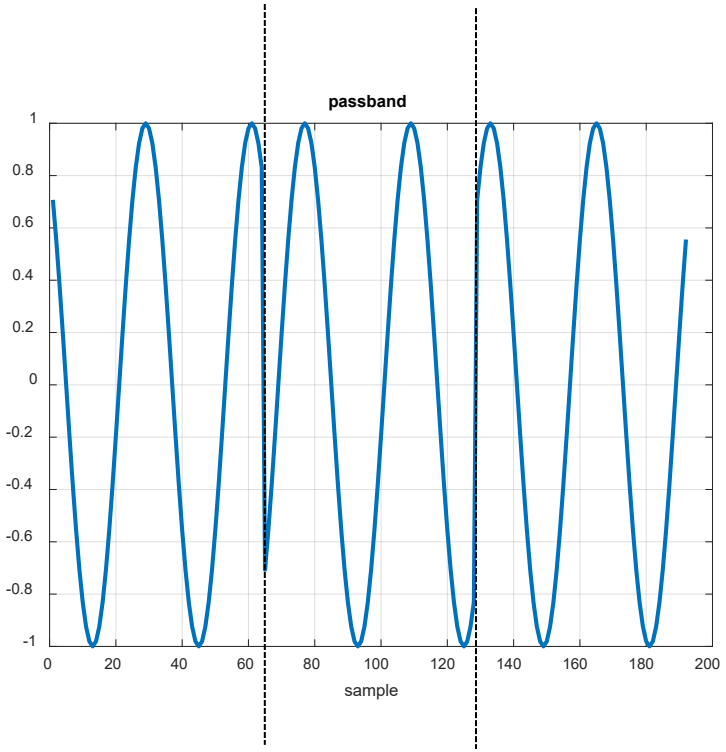


baseband

QPSK Signal

$$kT \leq t < (k+1)T$$

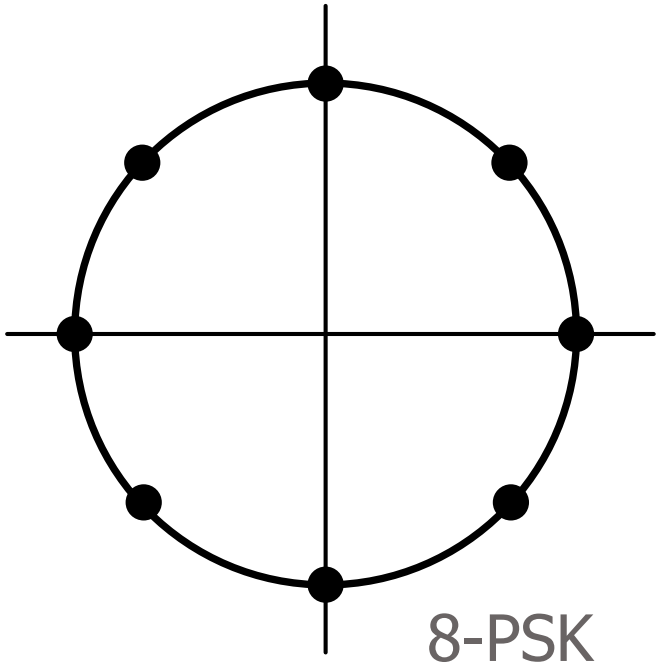




In the following, we will discuss baseband signal for modulation

M-ary PSK

- M-point constellation on a circle
- ($\cos(\theta + 2m\pi/M)$, $\sin(\theta + 2m\pi/M)$)



8-PSK

Differential Encoding

- In the receiver, there is often an ambiguity of the phase in the local carrier.
- Differential encoding is to map the information bits into the phase difference between two consecutive symbols instead of to the absolute phase of a symbol.
- Even with uncertainty of the absolute carrier phase, the phase difference between two consecutive symbols is preserved.

DQPSK/DEQPSK

- The in-phase and the quadrature-phase signal of the current symbol is determined by

$$I_k = I_{k-1}\cos(\phi_k) - Q_{k-1}\sin(\phi_k)$$

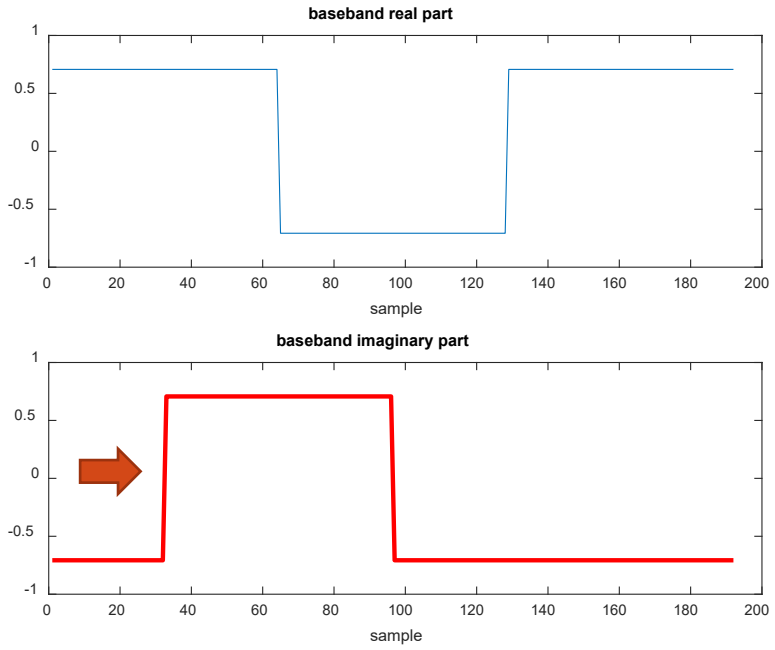
$$Q_k = I_{k-1}\sin(\phi_k) + Q_{k-1}\cos(\phi_k)$$

- The phase shift is limited to 0º, 90º, 180º, 270º.

Dibits	ϕ_k
00	0º
01	90º
10	270º
11	180º

OQPSK

- QPSK signals often have phase shift of 180º at symbol boundary, which causes the envelope go to zero when the signal is filtered.
- Offset QPSK (OQPSK) solves this problem by preventing the 180º phase shift through delaying $x_q(t)$ by $T/2$.

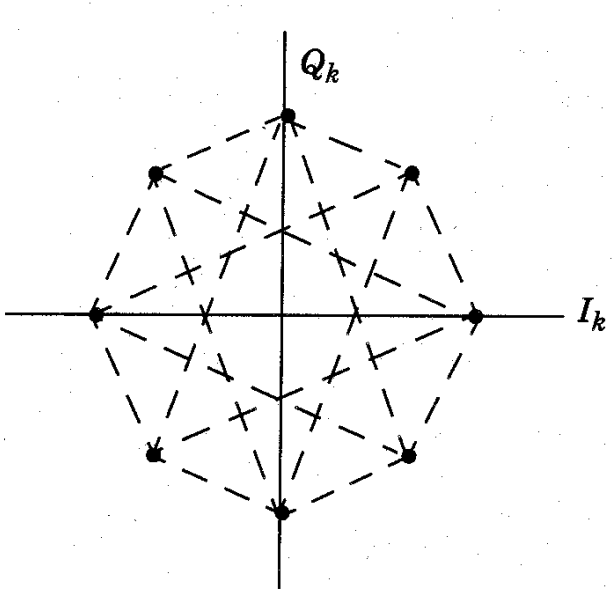


(π/4)-QPSK

- In (π/4)-QPSK, the maximum phase shift is limited to 135º, as compared to 180º in QPSK and 90º in OQPSK.
- There are two constellations of four signal points each.

(π/4)-QPSK

- The two constellations are shifted by π/4.
- Even and odd symbols use signal points from two constellations alternatively. So there are at least π/4 phase shift and at most 3π/4 phase shift.



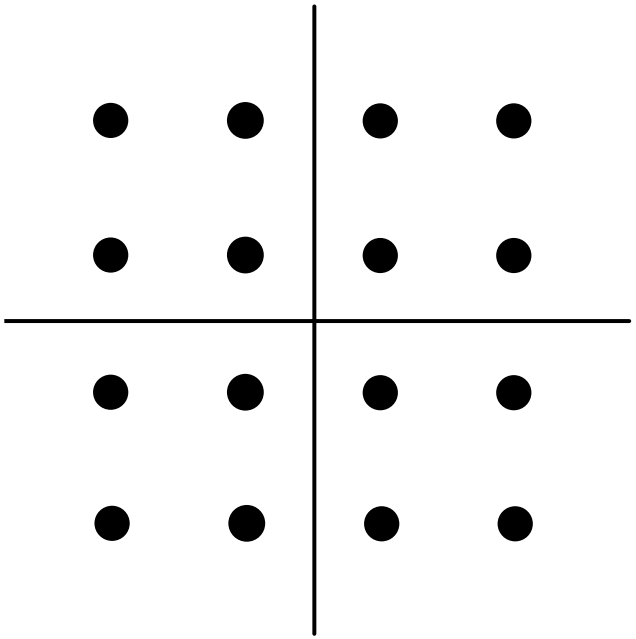
$(\pi/4)$-DQPSK

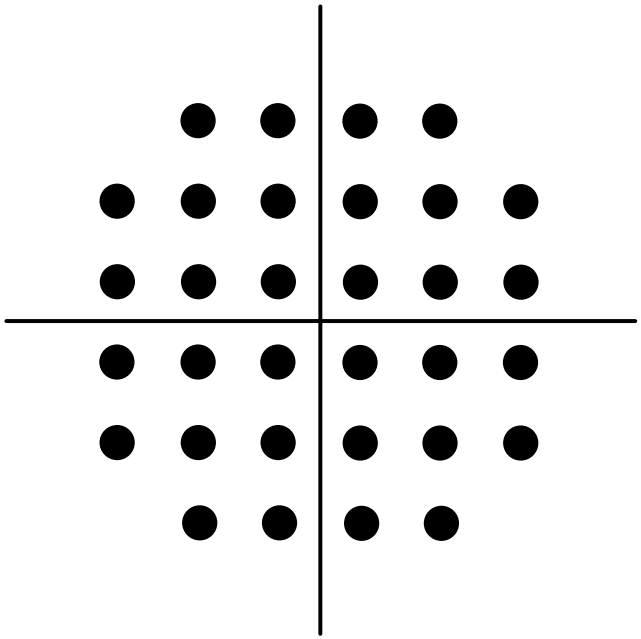
- Adopted by IS-54, PACS, and PHS
- With differential encoding, the $(\pi/4)$-QPSK is called $(\pi/4)$-DQPSK.
- The dibit to phase shift mapping is

Dibits	ϕ_k
00	$-3\pi/4$
01	$3\pi/4$
10	$-\pi/4$
11	$\pi/4$

Quadrature Amplitude Modulation (QAM)

- Higher-order modulation with higher spectral efficiency.
- Use both I- and Q-channel to transmit two baseband data streams.

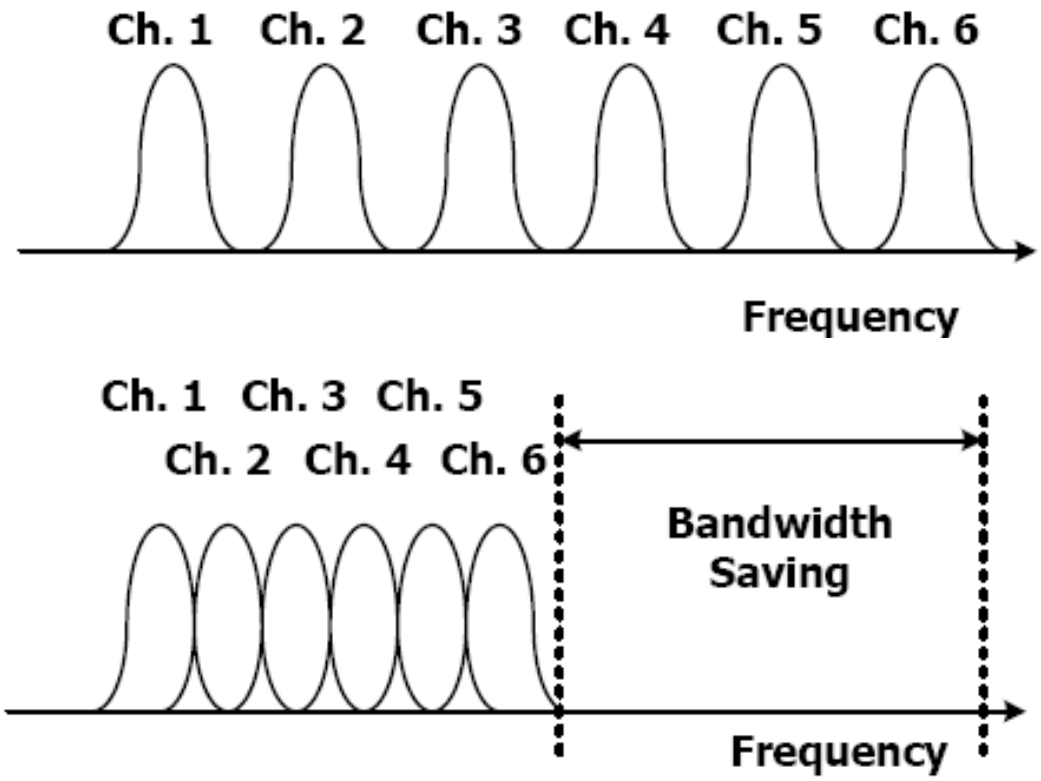




Multi-Carrier Modulation

Multi-Carrier Modulation

- FDM v.s. OFDM



Orthogonal Frequency Division Multiplexing (1/2)

- Continuous-time transmitted signal

$$x_{BB}(t) = \sum_{k=0}^{N-1} X_k e^{j2\pi f_k t}$$

- Discrete-time transmitted signal with sample period T_s

$$x_{BB}(nT_s) = \sum_{k=0}^{N-1} X_k e^{j2\pi f_k nT_s}$$

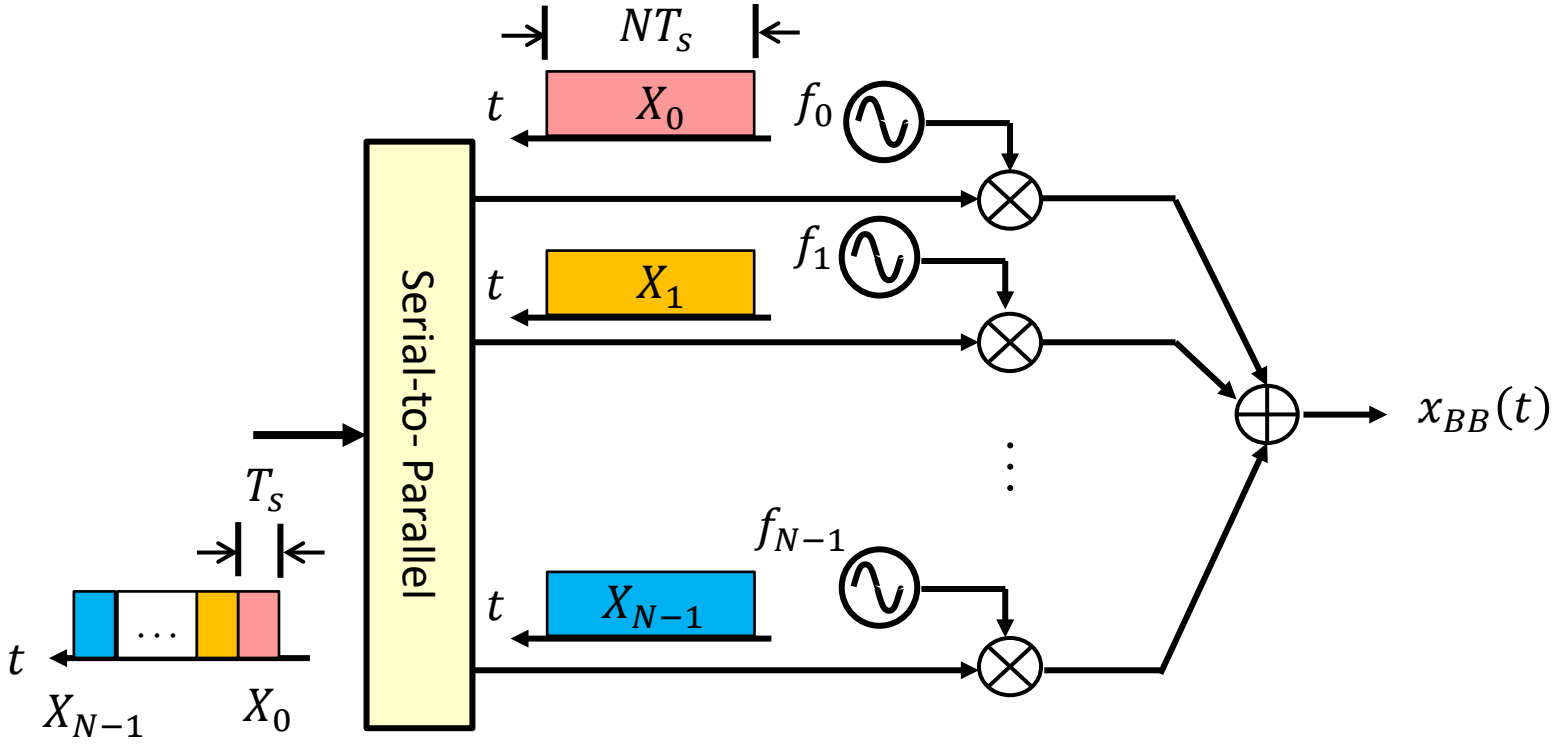
- If subcarriers are uniformly distributed with a spacing f_{sub}

$$x_{BB}(nT_s) = \sum_{k=0}^{N-1} X_k e^{j2\pi k f_{sub} nT_s}$$

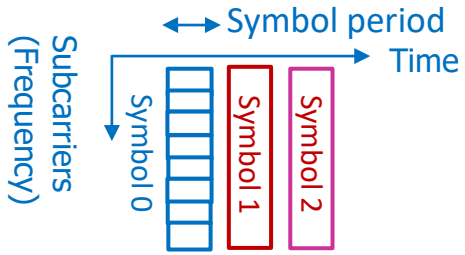
- The minimum separation to keep orthogonality $f_{sub} = 1/(NT_s)$

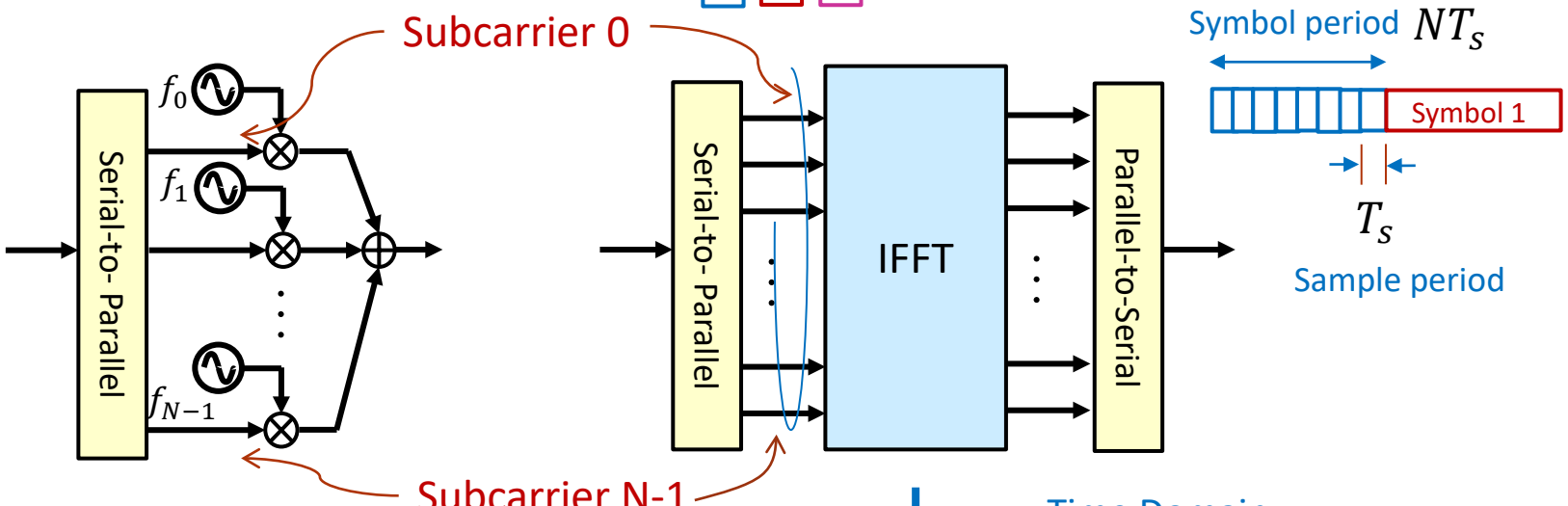
$$x_{BB}(nT_s) = \sum_{k=0}^{N-1} X_k e^{j2\pi kn/N}$$

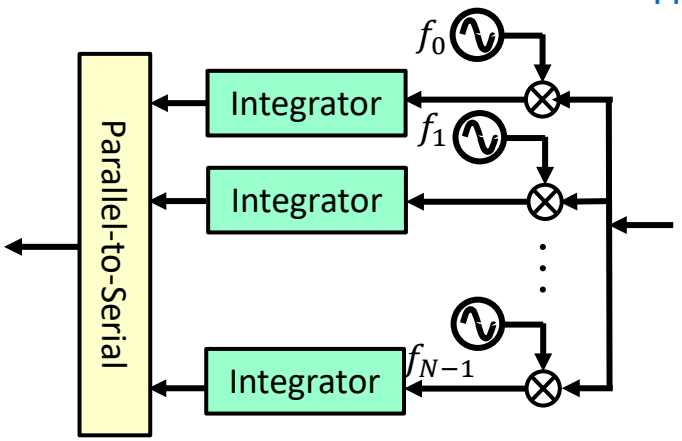
Orthogonal Frequency Division Multiplexing (2/2)



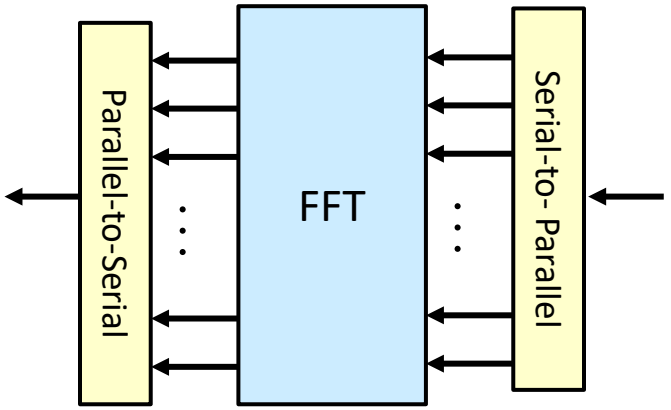
IFFT and FFT







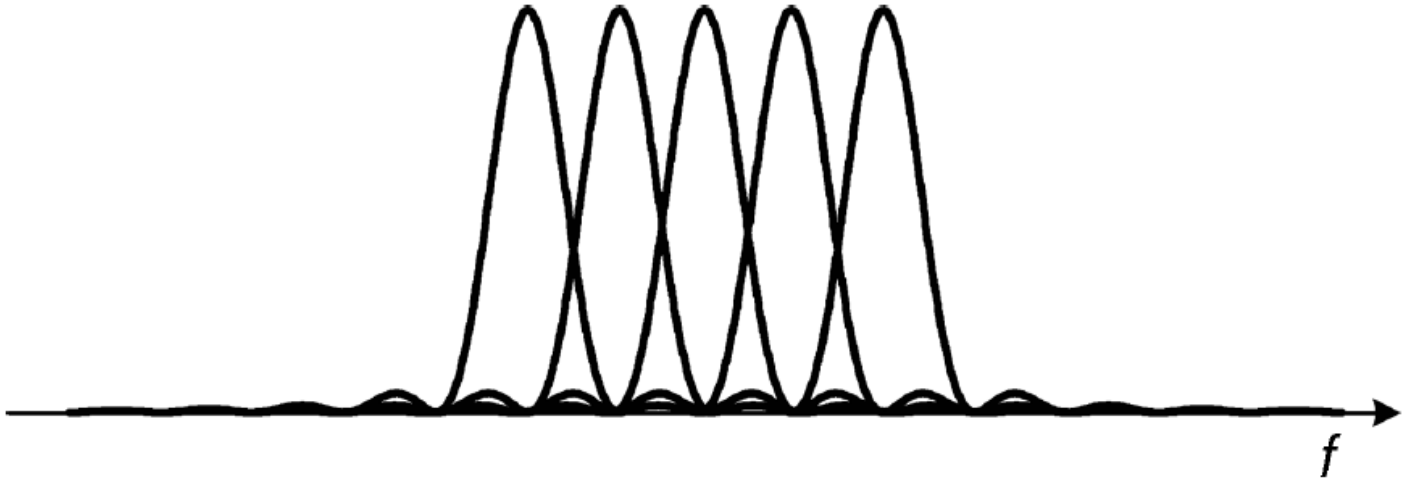
Continuous time



Discrete time

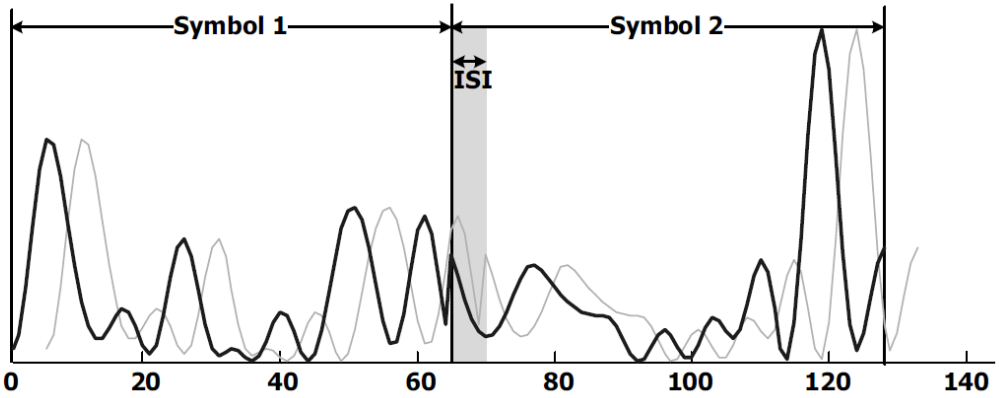
OFDM Power Spectrum Density

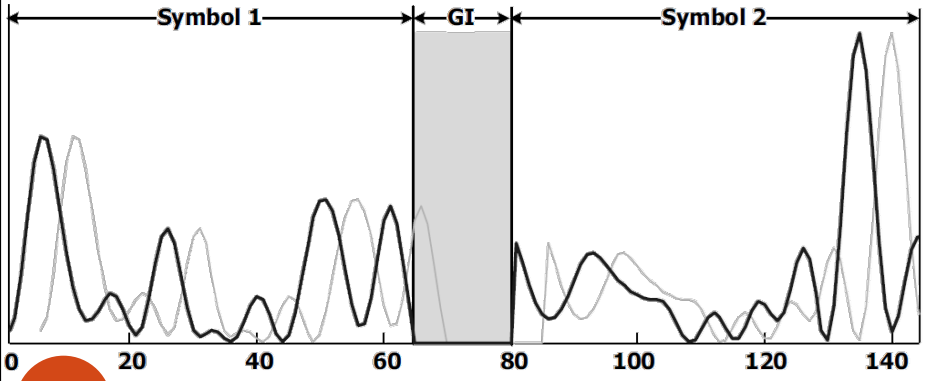
- Without specific time-domain windowing, the subcarriers have sinc-shaped spectra

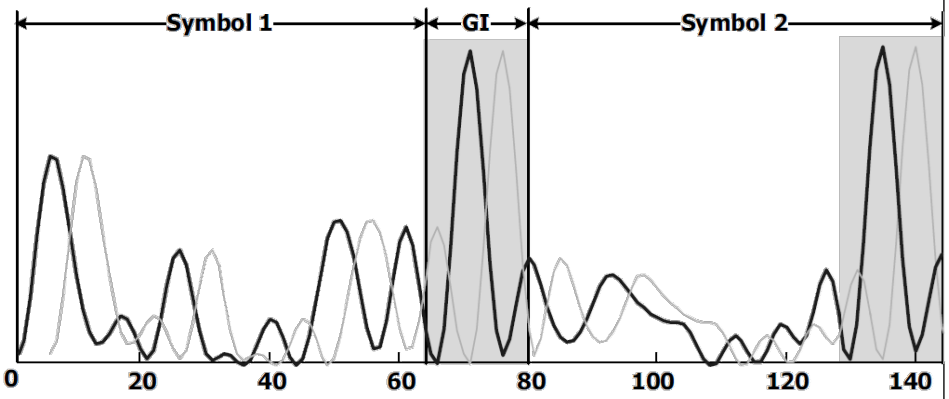


Guard Interval

- Zero-padding v.s. Cyclic prefix
- Usually CP is used for every symbol

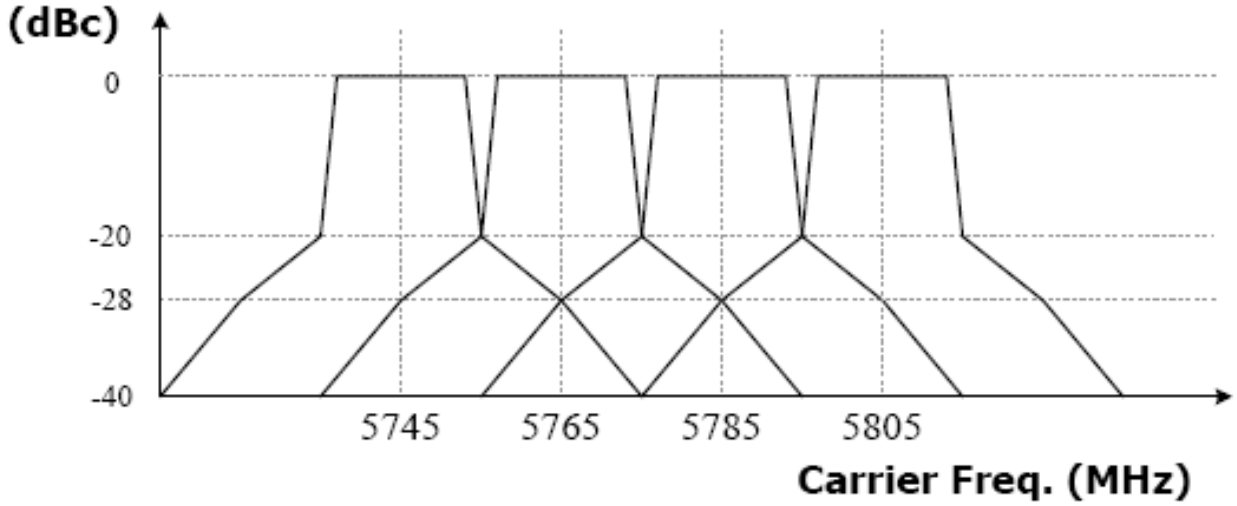






Null Subcarriers

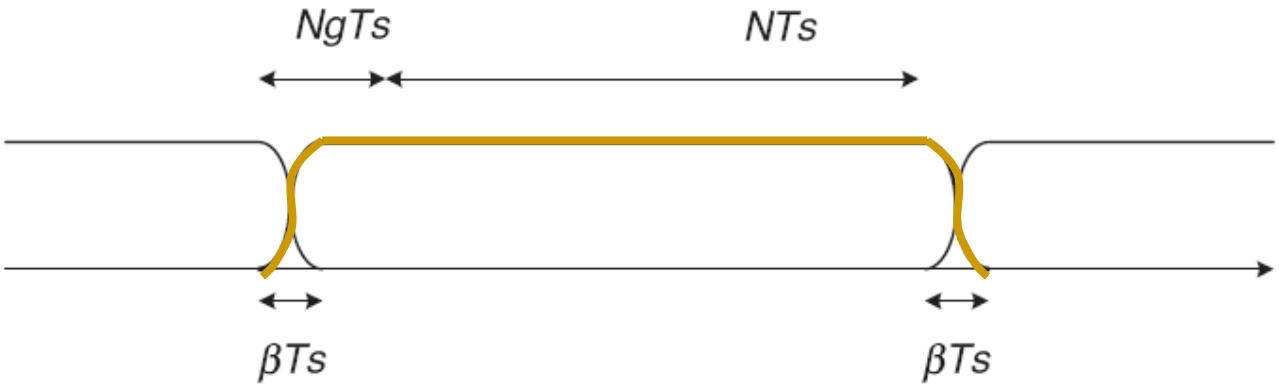
- DC subcarriers
  - Removal of unwanted DC component
- Guard subcarriers
  - Spectrum mask



Time-Domain Windowing

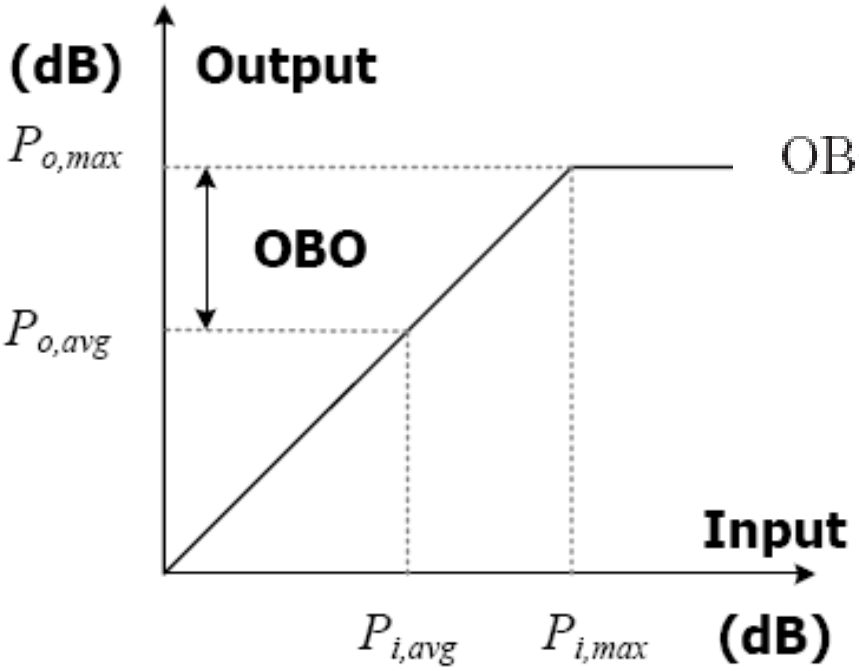
- To suppress the discontinuity between symbol boundary

$$w(t) = \begin{cases} 0.5 + 0.5\cos(\pi + \pi(t + N_g T_s)/(\beta T_s)) & -N_g T_s \le t \le (-N_g + \beta)T_s \\ 1 & (-N_g + \beta)T_s \le t \le NT_s, \\ 0.5 + 0.5\cos((t - NT_s)\pi/(\beta T_s)) & NT_s \le t \le (N + \beta)T_s \end{cases}$$



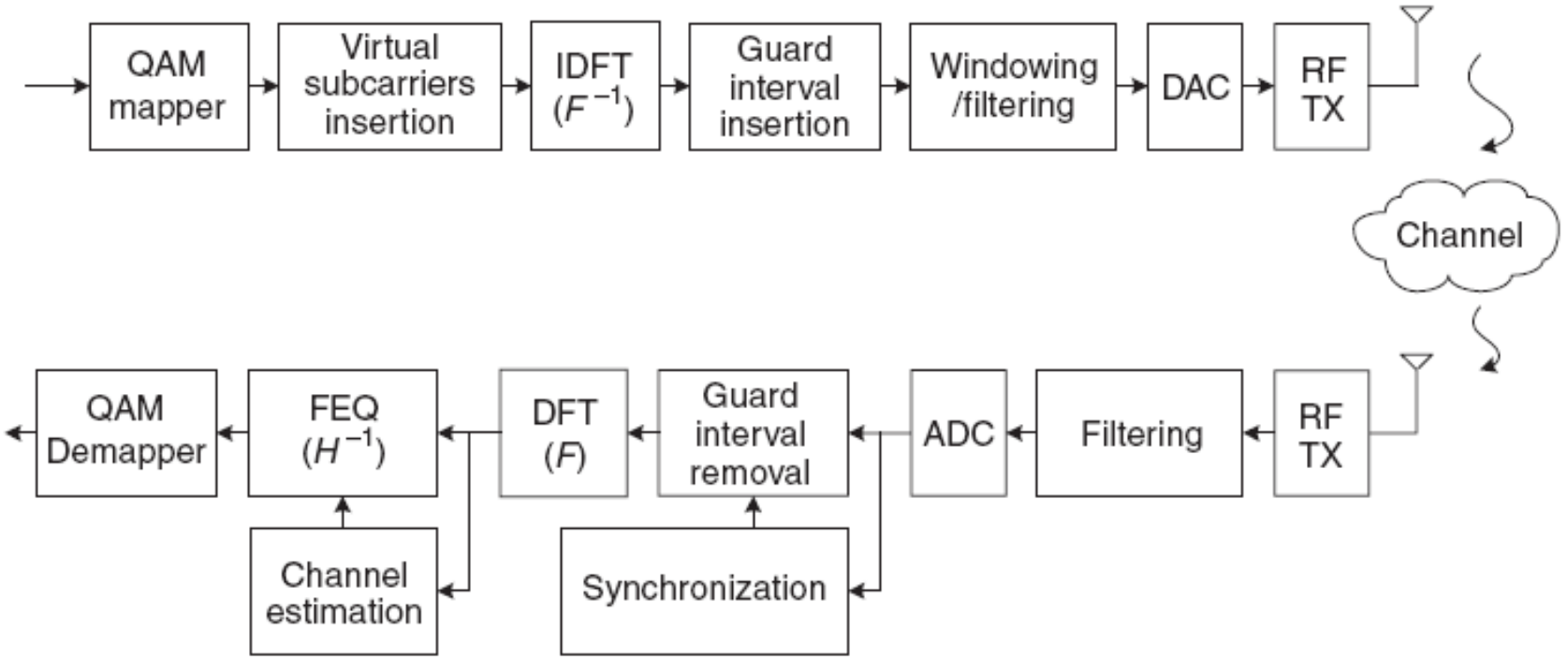
Peak-to-Average Power Ratio

- Large PAPR requires large output backoff point

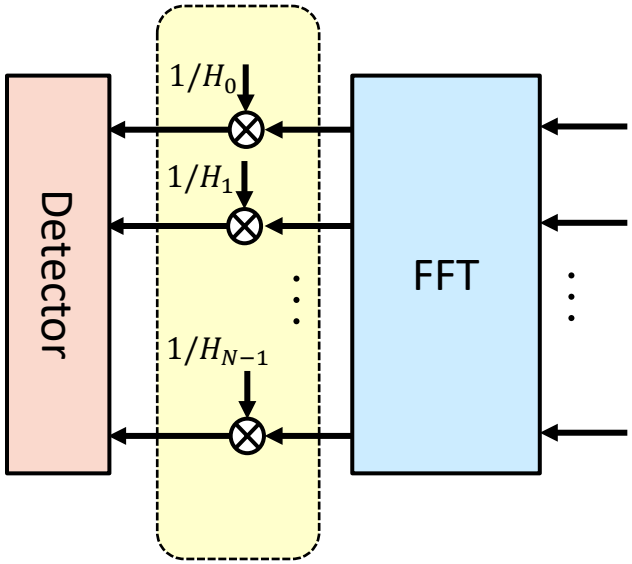


$$\mathrm{OBO} = 10\log_{10}\frac{P_{o,max}}{P_{o,avg}} \quad (\mathrm{dB}).$$

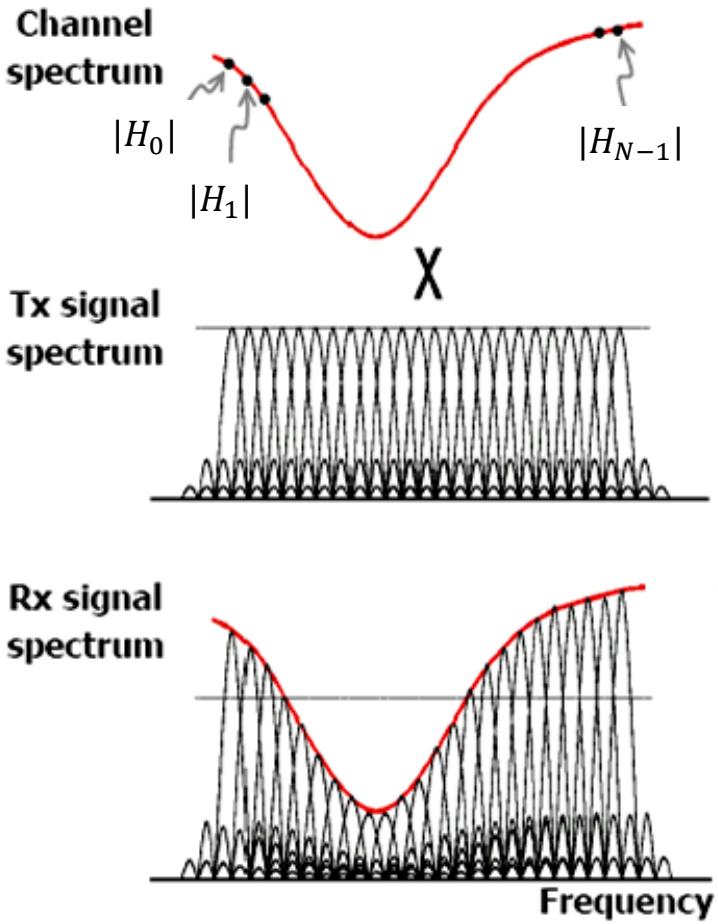
Typical Transceiver Architecture



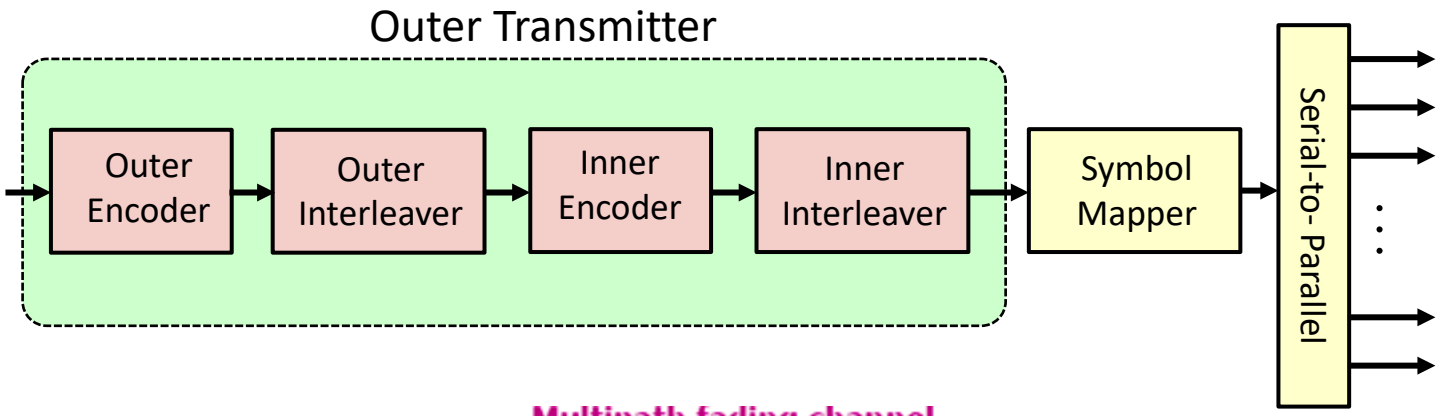
One-Tap Equalizer

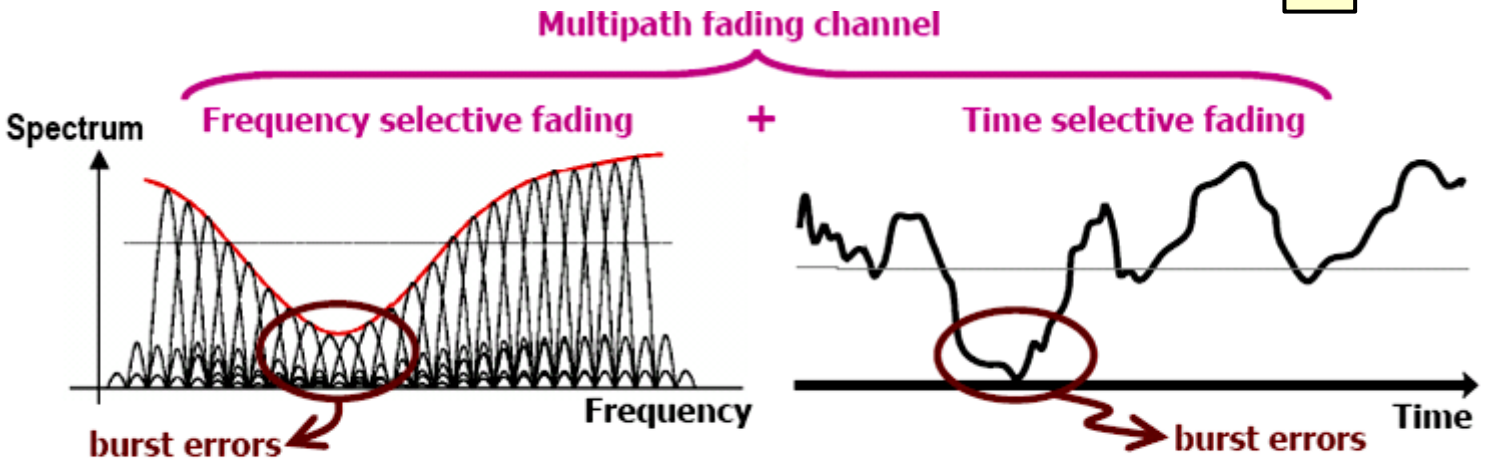


One-tap Equalizer



Coded OFDM (COFDM)





OFDM + Frequency/time interleaving + Powerful channel coding

System Parameters

- Subcarrier spacing v.s. coherence bandwidth
- Symbol time v.s. coherence time
- FFT size
- Guard interval
- Data rate

LTE Communication Standards

Transmission BW	1.25 MHz	2.5 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Sub-frame Duration	0.5 ms					
Subcarrier Spacing	15 kHz					
Sampling Frequency	1.92 MHz (1/2x3.84 MHz)	3.84 MHz	7.68 MHz (2x3.84 MHz)	15.36 MHz (2x3.84 MHz)	23.04 MHz (6x3.84 MHz)	30.72 MHz (8x3.84 MHz)
FFT Size	128	256	512	1024	1536	2048
Number of Occupied Subcarriers	76	151	301	601	901	1201
Number of OFDM Symbols Per Slot (Short/Long CP)	7/6					
CP Length (µs/samples)	(4.69/9)x6 (5.21/10)x1	(4.69/18)x6 (5.21/20)x1	(4.69/36)x6 (5.21/40)x1	(4.69/72)x6 (5.21/80)x1	(4.69/108)x6 (5.21/120)x1	(4.69/144)x6 (5.21/160)x1
	(16.67/32)	(16.67/64)	(16.67/128)	(16.67/256)	(16.67/384)	(16.67/512)

WiFi Communication Standards

- 802.11a/g

System Parameters	
Bandwidth (MHz)	20
FFT Size	64
Modulation	OFDM
Constellations	QPSK, 16-QAM, 64-QAM
Data Subcarriers	48
Subcarrier Spacing (KHz)	312.5
FFT Period (us)	3.2
Guard Interval Ratio	1/4
OFDM Symbol Period (us)	4.0
Data Rate (Mbps)	54

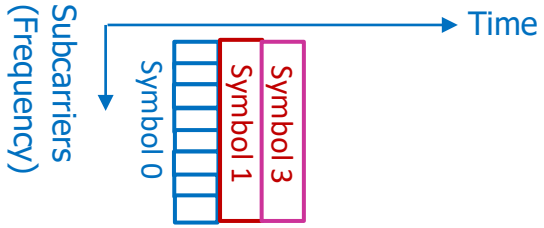
- 802.11n

System Parametersc		
Bandwidth (MHz)	20	**40**
FFT Size	64	**128**
Modulation	OFDM	
Constellations	QPSK, 16-QAM, 64-QAM	
Data Subcarriers	**52**	**108**
Subcarrier Spacing (KHz)	312.5	
FFT Period (us)	3.2	
Min. Guard Interval Ratio	1/4, **1/8**	
OFDM Symbol Period (us)	4.0, **3.6**	
Max. Data Rate (Mbps)	**288.9**	**600**
Antenna Configuration	**Up to 4**	

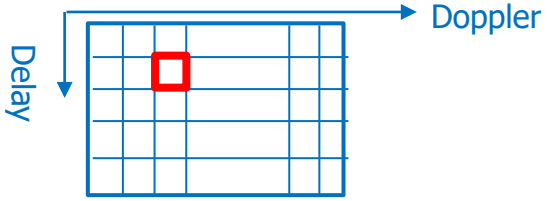
Orthogonal Time-Frequency Space Modulation (OTFS)

- OFDM system is vulnerable to time-varying channels and thus is not suitable for high-mobility scenarios.
- Doppler effect caused by high-mobility destroys the orthogonality among subcarriers.

- OTFS can resist time-varying channels
  - Two-dimensional modulation
    - Change the time-frequency domain to delay-Doppler domain



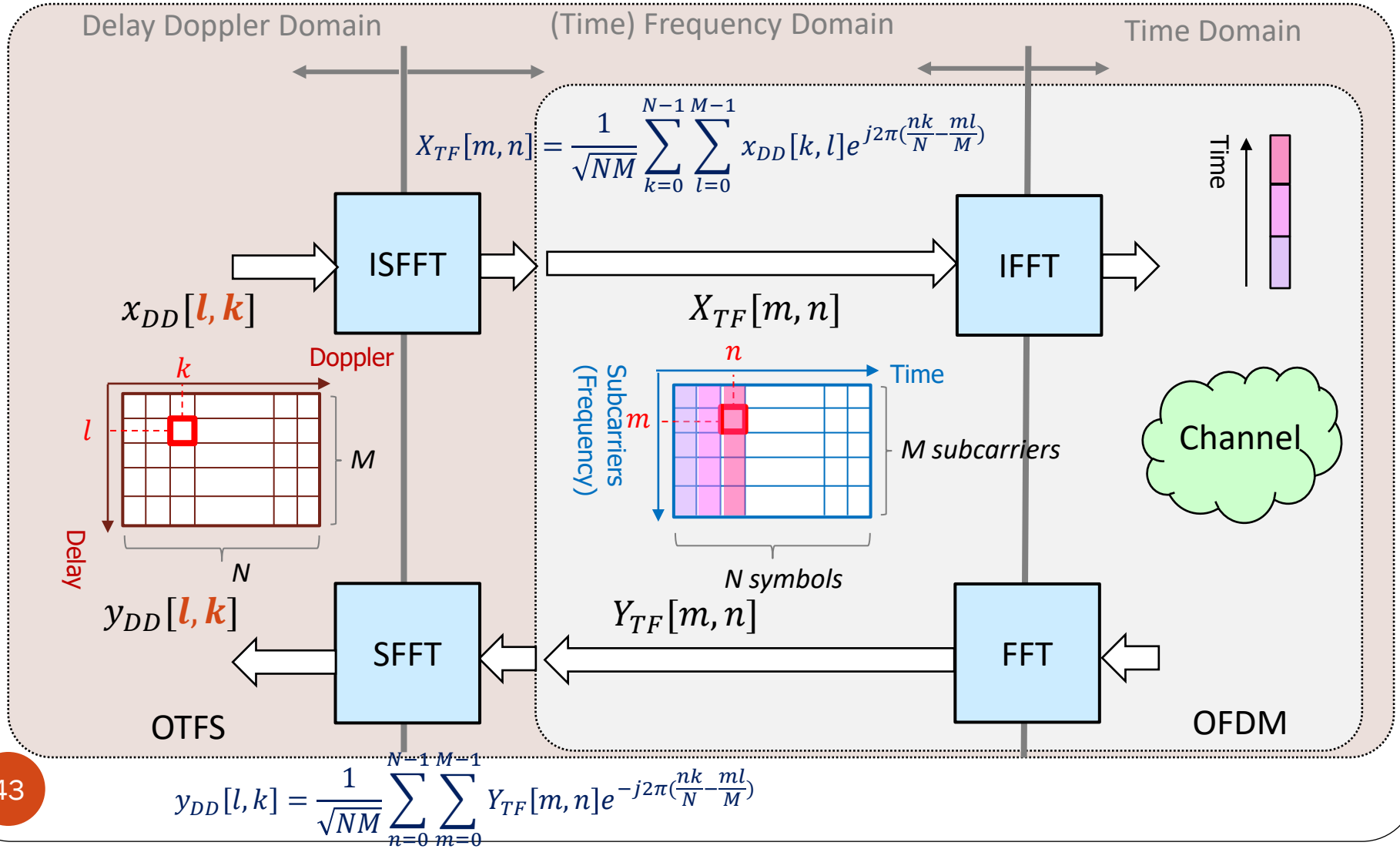
OFDM



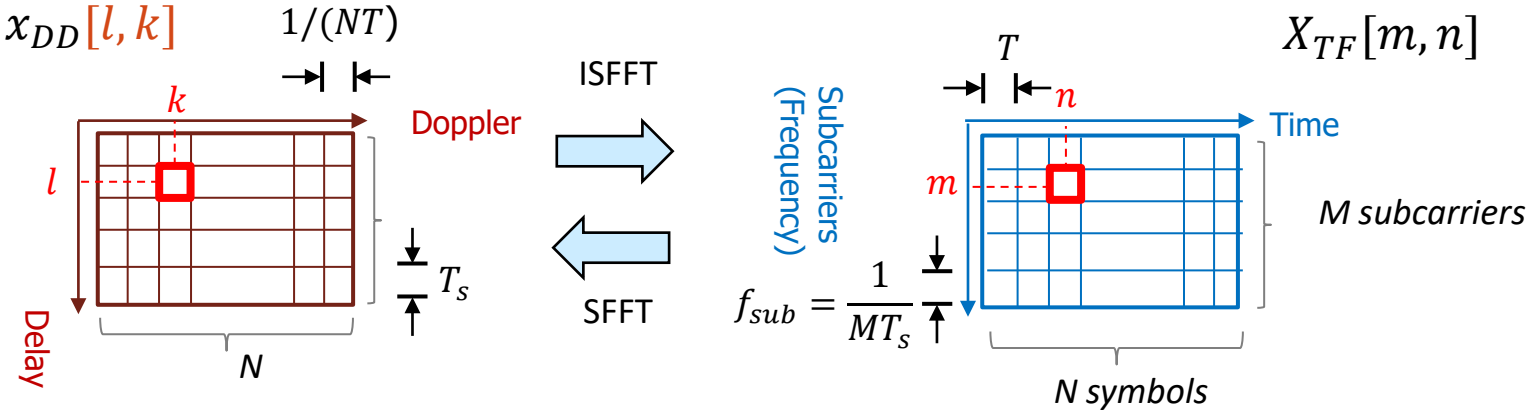
OTFS

OTFS Modulation

ISFFT: Inverse Symplectic FFT
SFFT: Symplectic FFT



OTFS Signal (1/2)



$$X_{TF}[m,n] = \frac{1}{\sqrt{NM}} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x_{DD}[l,k] e^{j2\pi(\frac{nk}{N} - \frac{ml}{M})}$$

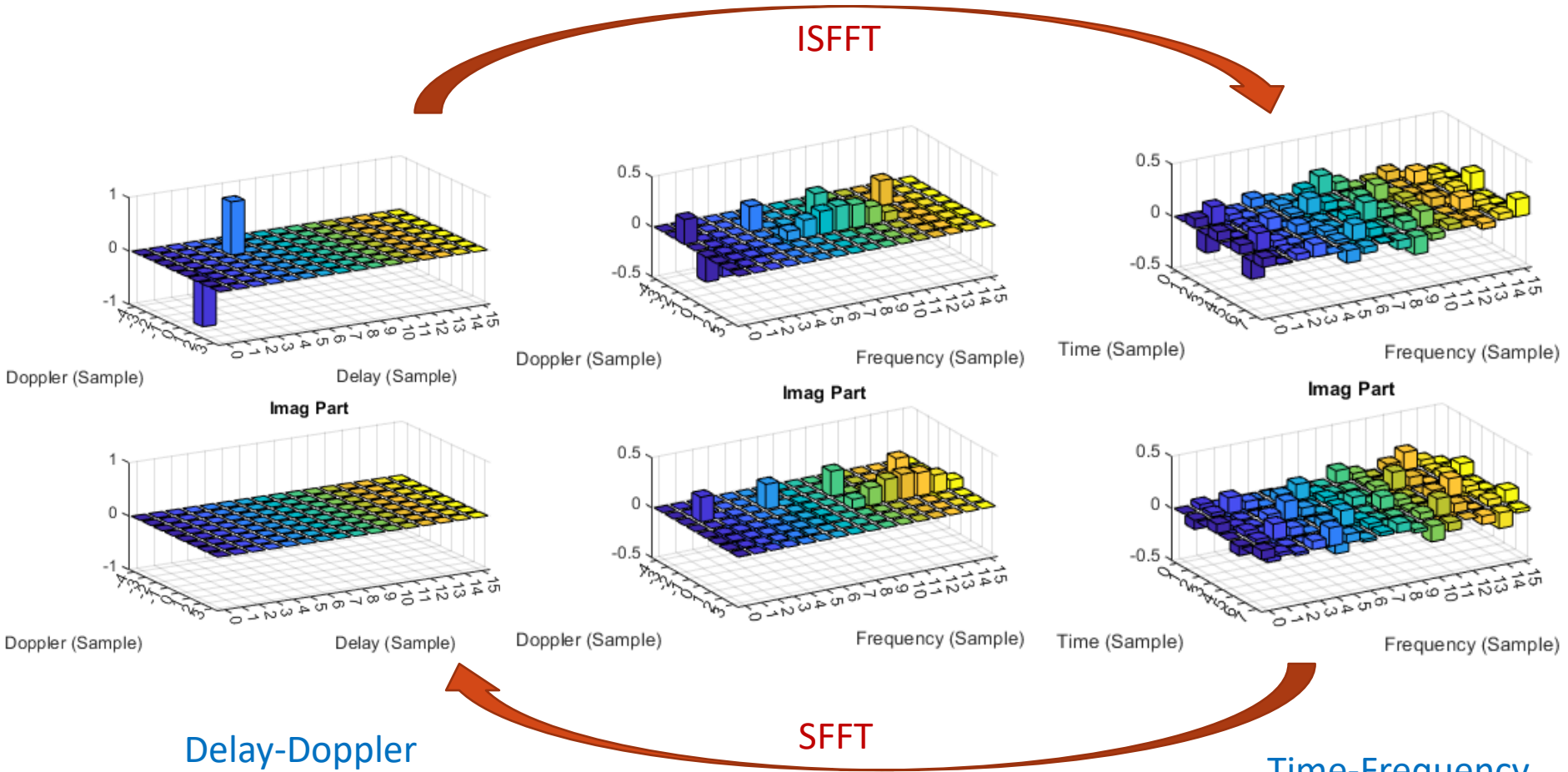
$$y_{DD}[l,k] = \frac{1}{\sqrt{NM}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y_{TF}[m,n] e^{-j2\pi(\frac{nk}{N} - \frac{ml}{M})}$$

From relationship between DFT and IDFT

Time Span NT versus Doppler Spectrum Resolution $1/(NT)$

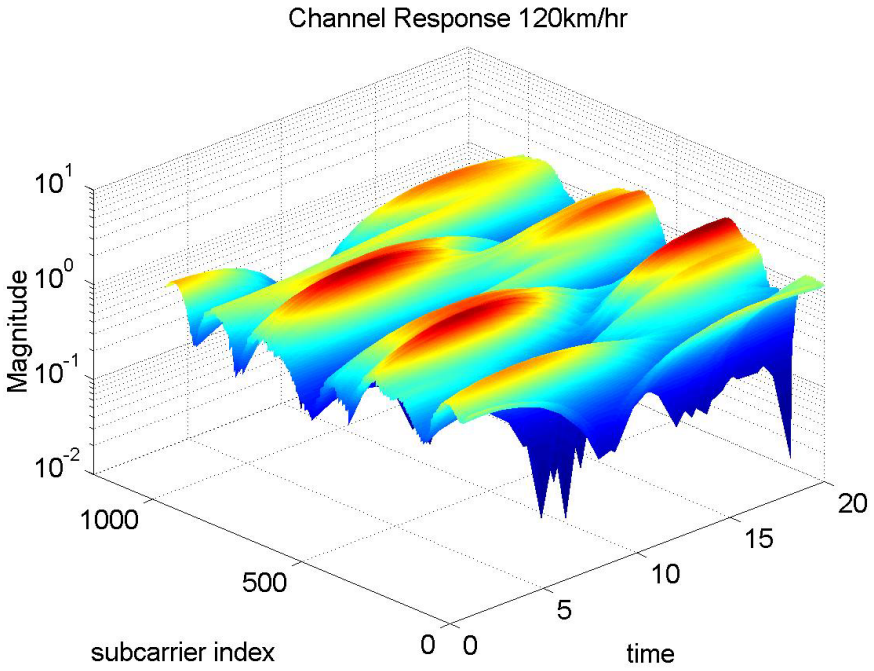
Sampling Period (Time Spacing) T_s versus Frequency Bandwidth $1/T_s$

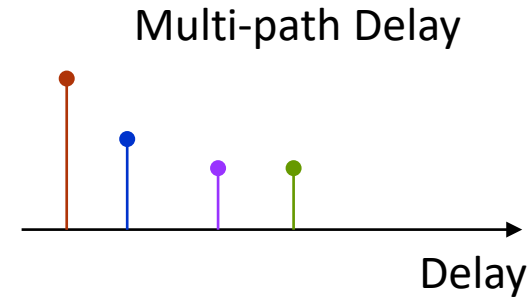
OTFS Signal (2/2)

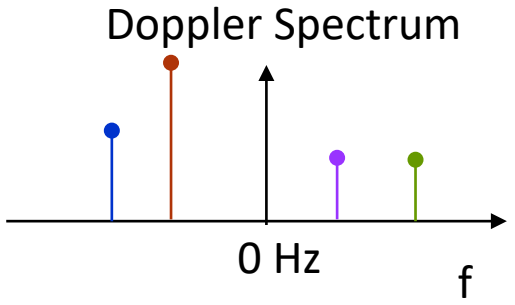


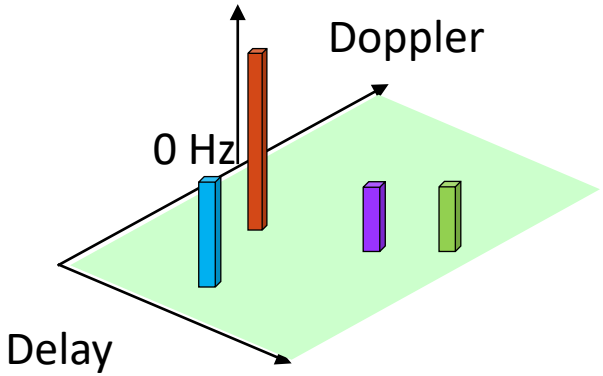
Why Delay Doppler?

- With time-varying channel in the time-frequency plane



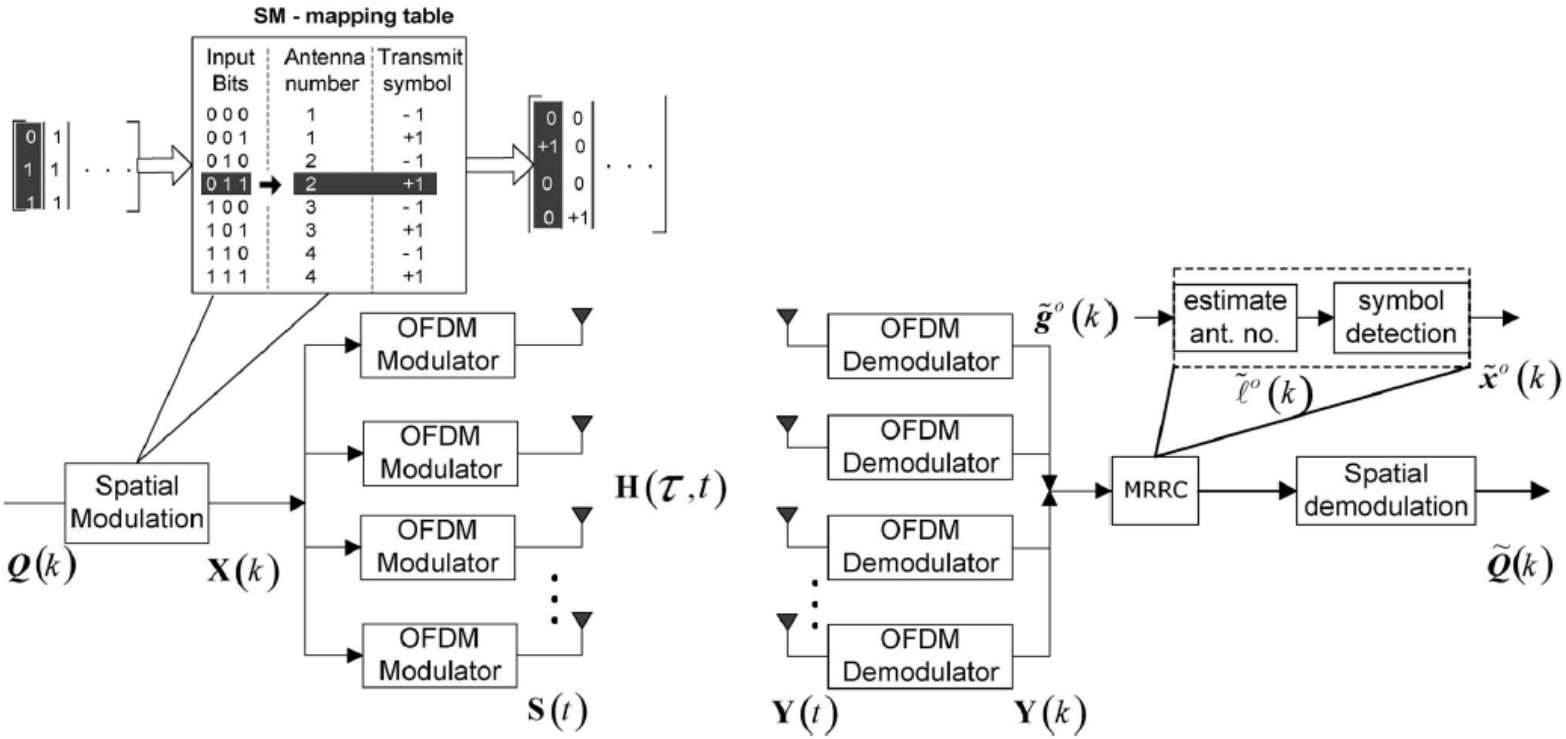






Spatial Modulation (1/2)

- Use different antennas to distinguish the data
- Generalized as Index Modulation



R. Y. Mesleh and et al. ``Spatial Modulation,'' IEEE Trans. Vehicular Technology, pp. 2228-2241, 2008.

Spatial Modulation (2/2)

- For M_T transmit antennas, if M -QAM is used
  - The number of bits that can be transmitted

$$\bar{m} = log_2(M_T) + log_2(M)$$

  - Constellation diagram and the number of transmit antennas can be traded-off for any number of transmitted information bits.

Input bits	N_t=2, M=4		N_t=4, M=2	
	Antenna number	Transmit symbol	Antenna number	Transmit symbol
000	1	+1+j	1	-1
001	1	-1+j	1	+1
010	1	-1-j	2	-1
011	1	+1-j	2	+1
100	2	+1+j	3	-1
101	2	-1+j	3	+1
110	2	-1-j	4	-1
111	2	+1-j	4	+1

R. Y. Mesleh and et al. ``Spatial Modulation,'' IEEE Trans. Vehicular Technology, pp. 2228-2241, 2008.